

Belief Dynamics in Time-varying Influence Networks with Logic Constraints and Stochastic Interactions

Chuafa Vachoima

8 May 2026

Project Supervisor: Dr Mengbin Ye

Abstract

Opinion formation in society emerges from complex interactions involving interpersonal influence, logical dependencies among topics, and evolving social connectivity. Belief systems consist of interdependent ideas and attitudes across multiple topics, and their evolution is shaped by social interactions among individuals. This study investigates belief dynamics in time-varying social influence networks with logic constraints and stochastic interactions.

The approach adopts an agent-based modelling framework that integrates the classical DeGroot model and Hegselmann–Krause model with logic constraints and stochastic interactions. The model introduces a novel influence-weight function that combines homophily-based bounded confidence with probabilistic interactions, enabling the social influence network to evolve dynamically over time. Opinions, logic structures, and social influence networks are represented in matrix form and implemented in MATLAB for numerical simulation and analysis.

The simulation results show that homophily plays a central role in the evolution of the social influence network. In particular, when the bounded confidence threshold exceeds a minimum value, the network gradually evolves into a strongly connected structure. The results also demonstrate that stochastic interactions generate oscillatory behaviour in opinion trajectories, while the logic structure determines whether opinions converge or remain diverse in strongly connected networks. Smaller bounded confidence thresholds lead to fragmented networks and stronger diversity in steady-state opinions.

1 Introduction

1.1 Belief System

Human beings are social animal and, at the same time, rational individuals with distinct beliefs (Usó-Doménech & Nescolarde-Selva, 2015). These beliefs arise from prior knowledge and experience. Human sociability contributes to the formation of belief systems. Belief systems can be described as configurations of ideas and attitudes across various domains, such as religion, politics, worldview, and morality, where the elements are bound together through functional interdependence (Converse, 2008). Each individual holds an initial belief or stance on a particular topic. In society, individuals interact with one another and form social networks through which their opinions continuously influence each other. Through these interactions, beliefs are not static; rather, they evolve over time as individuals update their opinions based on the influence of others.

The study of how beliefs evolve through social interactions has become an important topic in opinion dynamics. Decisions, whether individual or collective, are often shaped by these beliefs, which in turn influence how people behave and approach problems. As individuals interact within a network, the outcomes of the opinions depend on the structure of the network and the mechanisms governing social influence. Empirical studies have shown that public opinion on social issues, such as gay marriage and gender roles, are not static but instead fluctuate over time (León-Medina, 2019). Similar phenomena can also be observed in aggregate voting patterns in politics (Kramer, 1971). Such collective opinions play a critical role in shaping political decisions and public policies which have a direct impact on society.

Understanding how collective belief patterns emerge from individual interactions is therefore crucial for explaining social phenomena such as consensus formation, persistent opinion fluctuations, clustering and opinion fragmentation. Such understanding provides valuable insights into how opinions in society evolve and can support more informed decision-making in areas such as policymaking, governance, and social planning.

1.2 Literature Review

To better understand the mechanisms underlying opinion evolution, agent-based models have been developed to explain how individuals adjust their views in response to social influence and interactions. The classical DeGroot model represents opinion updating as a weighted averaging process over an agent's own opinion and those of their neighbours (Degroot, 1974). Real-world opinion formation, however, is substantially more com-

plex, motivating numerous extensions of this foundational framework to better capture the complexities of social influence and belief evolution.

The Friedkin–Johnsen (FJ) model incorporates stubbornness by allowing individuals to partially retain their initial opinions (Friedkin et al., 2016). Just like the DeGroot model, the FJ model assumes a static social influence network, this assumption may not be true in reality, as relationships and social influence often evolve over time. The Hegselmann–Krause (HK) model addresses this limitation by introducing bounded confidence, a mechanism that captures homophily, where individuals are only influenced by neighbours with sufficiently similar opinions, allowing the social influence network to evolve dynamically (Hegselmann & Krause, 2002).

Recent work has further expanded this framework. Cheng et al. (2025) generalised the HK model to multidimensional settings with heterogeneous confidence bounds and probabilistic random interactions, offering a more realistic depiction of social communication patterns. Brooks and Porters (2020) adapted the HK model for online social networks, incorporating media and non-media agents and modelling how content quality shape opinion propagation.

Beyond interpersonal influence, opinion formation often occurs across multiple logically interdependent issues. Ye et al. (2020) explored how such logical interdependency, captured by a logic matrix, influence final opinion outcomes. They found that consensus between topics is achieved when the logic matrices are irreducible, while the reducible cascade interdependence relationship and heterogeneity among the logic matrices of individuals can lead to disagreement (Ye et al., 2020).

In real-world social contexts, social influence networks are rarely static and can evolve over time due to several factors, such as geographical locations, accessibility, availability of individuals, and temporal constraints. Consequently, the interactions among the individuals do not remain constant throughout the observation period. In such circumstances, incorporating randomness into the modelling framework provides a more realistic representation of how interactions occur over time, thereby improving the realism and accuracy of the simulation model. Furthermore, interactions among like-minded individuals tend to occur more frequently than those among individuals with dissimilar opinions (McPherson et al., 2001). Individuals evaluate others' opinions relative to their own before deciding whether to adopt them, and the extent of such adoption varies. In addition, discussions often involve multiple interdependent topics. These observations motivate the study of opinion evolution in time-varying social influence networks that incorporate homophily and heterogeneous adoption behaviour, where interactions oc-

cur stochastically and opinions are further shaped by logical interdependency among topics. Such a framework may provide deeper insights into complex phenomena such as persistent opinion fluctuations, clustering, fragmentation, and the emergence of diverse collective belief patterns.

2 Research Questions

Inspired by the Hegselmann–Krause model and the logic matrix framework proposed by Ye et al. (2020), this study aims to synthesise these ideas, introduce several modifications, and investigate the following research question:

How do time-varying social influence networks, incorporating logical constraints and stochastic interactions, affect the resulting opinion dynamics and long-term outcomes?

Further investigation will be conducted to address the following questions:

1. What types of social influence network structures emerge under homophily, where individuals are influenced only by neighbours whose opinions are sufficiently similar to their own, in the presence of random interactions?
2. How do variations in social influence network structures and logical constraints, both individually and jointly, shape the evolution of opinion dynamics over time?

The study reveals that homophily is the primary mechanism driving the formation of social influence network structures. A strongly connected social influence network emerges when the bounded-confidence threshold exceeds a minimum tolerance level. In contrast, weakly connected networks are the main source of opinion fragmentation, while local clustering may arise from strongly connected components embedded within a weakly connected network. Persistent fluctuations in opinion values are primarily explained by the incorporation of stochastic interactions, whereas logical interdependency structures play a crucial role in determining whether opinions converge to consensus or remain diverse within a strongly connected network. Overall, this study contributes to explaining why some social networks are well-connected while others are fragmented. It also provides insights into the fluctuations of opinions observed in politics and social issues that were not explained by other models, as well as the emergence of consensus and opinion diversity over time.

3 Methods

In this study, MATLAB is employed to conduct simulations of the evolution of opinions within the proposed opinion dynamics framework. The opinion update rule for agent i is given by

$$x_i(t+1) = \sum_{j=1}^n w_{ij}(t) C_i x_j(t), \quad i = 1, \dots, n \quad (1)$$

where $x_i(t) \in \mathbb{R}^m$ denotes the opinion vector of agent i on m topics at discrete time t , $t = 1, 2, \dots, T$. The opinion vector is represented as

$$x_i(t) = \begin{bmatrix} x_{1i}(t) \\ x_{2i}(t) \\ \vdots \\ x_{mi}(t) \end{bmatrix}$$

Here, $x_{mi}(t)$ denotes the opinion value of the i -th individual on the m -th topic at time t , where $x_{mi}(t) \in [-1, 1]$. The value -1 corresponds to disagreement, while 1 corresponds to agreement.

The logic matrix¹, $C_i \in \mathbb{R}^{m \times m}$ captures the interdependency among topics for agent i .

The weight of the influence from agent j to agent i at time t is represented by $w_{ij}(t)$, where $w_{ij}(t) \in [0, 1]$, with 0 denoting no influence and 1 denoting maximum influence.

The proposed framework assumes that $w_{ij}(t)$ is a decreasing linear function when the bounded confidence condition is satisfied, unlike the Hegselmann–Krause model which assigns equal influence weights for the agent's neighbours. The influence weights are defined as

$$w_{ij}(t) = \begin{cases} 1 - \frac{\|x_i(t) - x_j(t)\|}{d}, & \text{if } \|x_i(t) - x_j(t)\| \leq \epsilon \text{ and } j \neq i, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

In the influence network, when an individual evaluates the opinions of others, greater weight is assigned to agents whose opinions are closer to their own. Hence, this formulation implies that agents are more influenced by individuals whose opinions are more similar to theirs. The quantity $\|x_i(t) - x_j(t)\|$ denotes the Euclidean distance, which measures the similarity between multidimensional opinion vectors.

¹The detailed structure of C_i is described in Subsection 3.1.

The parameter d represents the maximum Euclidean distance between two extreme opposite opinion vectors. Since $x_{pi} \in [-1, 1]$, the maximum possible distance is given by

$$d = 2\sqrt{m}. \quad (3)$$

For simplicity, it is assumed that all agents share the same bounded confidence threshold ϵ . The diagonal entries at time t are generated as

$$w_{ii}(t) \sim \text{Uniform}(0, 1)$$

to prevent excessive self-reliance, since according to equation (2), an individual's own opinion is always the most similar to itself and would otherwise receive disproportionately large influence weights.

Finally, the influence weights are normalised to ensure that $W(t)$ is row-stochastic:

$$w_{ij}(t) = \frac{w_{ij}(t)}{\sum_{j=1}^n w_{ij}(t)}. \quad (4)$$

Figure 1 below illustrates how the influence weight decreases linearly with increasing opinion distance.

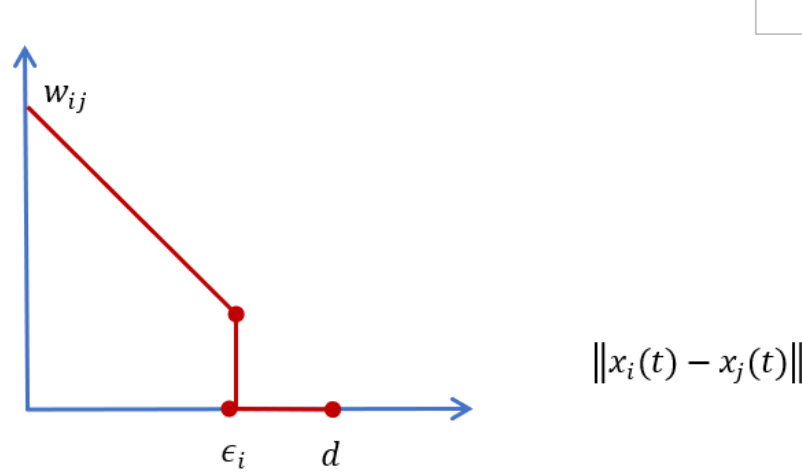


Figure 1: Influence weight function .

To capture a more realistic scenario, stochastic interactions are incorporated following the framework of Cheng et al. (2024). In this setting, each potential pair of agents (i, j) has a distinct probability of interaction at each time step. Even if an interaction occurs, agent i is only influenced by agent j when the bounded confidence condition in(2) is satisfied.

First, an access probability matrix P is initialised such that

$$P \in \mathbb{R}^{n \times n}, \quad P_{ij} \sim \text{Uniform}(\delta, 1),$$

where P_{ij} denotes the probability that agent i has access to the opinion of agent j . The parameter δ represents the minimum probability threshold, ensuring that every pair (i, j) has at least some chance of accessing each other's opinions.

At each time step t , for every pair (i, j) , a random variable is drawn as

$$r_{ij}(t) \sim \text{Uniform}(0, 1).$$

If the following condition holds, a one-way interaction from agent j to agent i occurs:

$$r_{ij}(t) < P_{ij}. \quad (5)$$

For computational implementation in MATLAB², the opinions of all agents, and the associated social influence network are represented in matrix form.

3.1 Matrix Representation of the Model

Matrix $X(t) \in \mathbb{R}^{m \times n}$ is constructed to collect the opinion values of n individuals across m topics at time t , as shown below:

$$X(t) = \begin{bmatrix} x_1(t) & x_2(t) & \cdots & x_n(t) \end{bmatrix} = \begin{bmatrix} x_{11}(t) & x_{12}(t) & \cdots & x_{1n}(t) \\ x_{21}(t) & x_{22}(t) & \cdots & x_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1}(t) & x_{m2}(t) & \cdots & x_{mn}(t) \end{bmatrix}$$

Here, $x_{mn,t}$ denotes the opinion value of the n -th individual on the m -th topic at time t .

Matrix $W(t) \in \mathbb{R}^{n \times n}$ is defined to represent the social influence network at time t :

$$W(t) = \begin{bmatrix} w_{11}(t) & w_{12}(t) & \cdots & w_{1j}(t) \\ w_{21}(t) & w_{22}(t) & \cdots & w_{2j}(t) \\ \vdots & \vdots & \ddots & \vdots \\ w_{i1}(t) & w_{i2}(t) & \cdots & w_{ij}(t) \end{bmatrix}$$

²The detailed implementation of the code is available at <https://github.com/Chuaafavang/Modelling-and-Simulation-of-Belief-System-Dynamics-on-Influence-Networks.git>

Here, $w_{ij}(t)$ represents the influence weight from individual j to individual i at time t . For instance, in Figure 2, w_{12} denotes the influence weight from individual 2 to individual 1, represented by the directed arrow pointing toward individual 1.

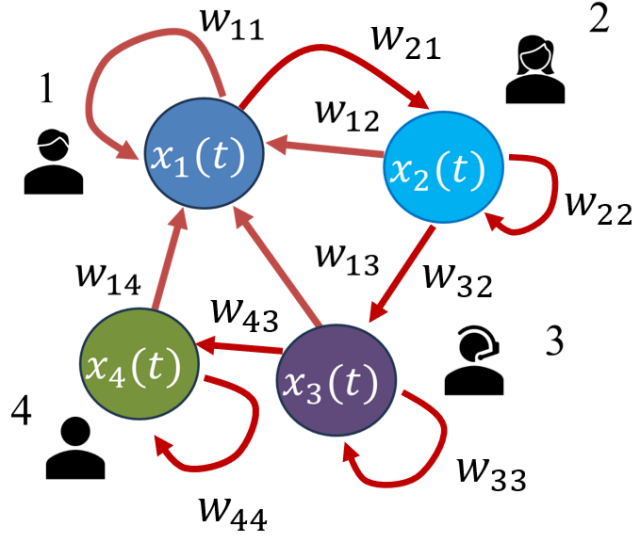


Figure 2: A strongly connected social influence network.

Two important characteristics of social influence networks are strongly connected influence networks and weakly connected influence networks. Figure 2 shows an example of a strongly connected influence network. In this type of network, every node is reachable from every other node by following the edges with respect to the direction of each edge (Ye et al., 2020). Hence, it is equivalent to every individual has direct or indirect influence of each other. The W matrix to represent the social influence network in Figure 2 is defined as:

$$W = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} \\ w_{21} & w_{22} & 0 & 0 \\ 0 & w_{32} & w_{33} & 0 \\ 0 & 0 & w_{43} & w_{44} \end{bmatrix}$$

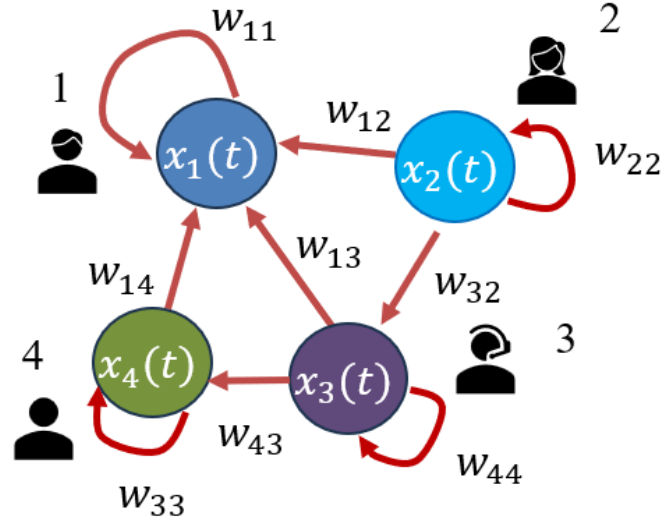


Figure 3: A weakly connected social influence network.

Figure 3 shows a weakly connected influence network. In this type of network, there exists at least one node that is not reachable by any nodes. Node 2 is not reachable by any nodes in Figure 3. Therefore, there exists at least one individual who does not receive any influence from anyone. The W matrix to represent the social influence network in Figure 3 is defined as:

$$W = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} \\ 0 & w_{22} & 0 & 0 \\ 0 & w_{32} & w_{33} & 0 \\ 0 & 0 & w_{43} & w_{44} \end{bmatrix}$$

Matrix C_i is introduced to capture the logical interdependency among topics for individual i .

$$C_i = \begin{bmatrix} c_{11,i} & c_{12,i} & \cdots & c_{1q,i} \\ c_{21,i} & c_{22,i} & \cdots & c_{2q,i} \\ \vdots & \vdots & \ddots & \vdots \\ c_{p1,i} & c_{p2,i} & \cdots & c_{pq,i} \end{bmatrix}$$

Here, $c_{pq,i}$ denotes the weight of logical dependence from topic q to topic p for individual i , where $c_{pq,i} \in [-1, 1]$, with positive values representing supportive logical dependence and negative values representing contradictory dependence. $p, q = 1, \dots, m$.

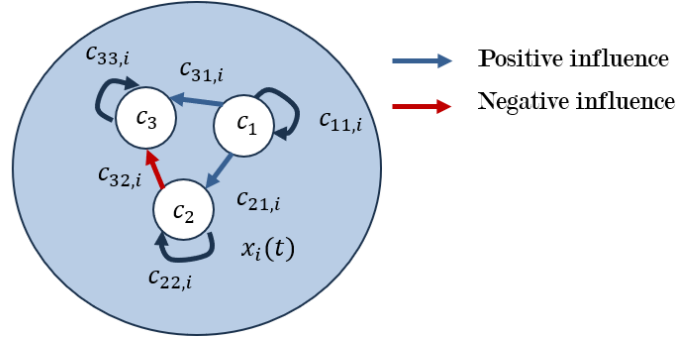


Figure 4: A logical interdependence of 3 topics within an individual.

The matrix C_i of an individual i in Figure 4 is defined as:

$$C_i = \begin{bmatrix} c_{11,i} & 0 & 0 \\ c_{21,i} & c_{22,i} & 0 \\ c_{31,i} & -c_{32,i} & c_{33,i} \end{bmatrix}$$

The matrix C_i represents the logical interdependency among the 3 topics. Topic c_1 depends only on itself and therefore has a single non-zero entry $c_{11,i}$. Topic c_2 is positively influenced by topic c_1 , indicating that opinion 1 supports opinion 2. The negative entry $-c_{32,i}$ implies that topic c_3 is contradicted by topic c_2 .

3.2 Model Initialization and Network Formation Mechanism

3.2.1 Initial Opinion Generation

The initial opinions are generated as

$$x_{pi}(0) \sim \text{Beta}(\alpha, \beta), \quad p \in \{1, \dots, m\}, \quad i \in \{1, \dots, n\},$$

with

$$x_{pi}(0) \in [-1, 1].$$

This specification allows each topic p to have a different mean and spread, thereby ensuring diversity in the initial opinion values across agents. The value -1 represents complete disagreement, whereas 1 represents complete agreement. The simulations consider different population sizes, namely $n = 6$ (small-scale), $n = 30$ and $n = 100$ (medium-scale), and $n = 1000$ (large-scale). The case $n = 1000$ is included to provide better and more accurate insight into the steady-state distribution of opinion values in a large-scale population. Figure 5 illustrates an example of the initial opinion distribution used in the simulation experiments for $n = 100$.

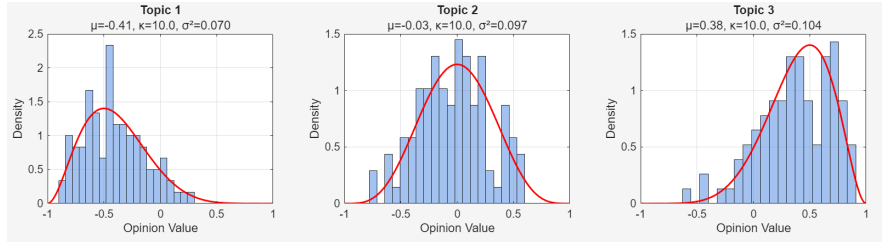


Figure 5: The initial opinion distribution across agents and topics.

The blue histograms represent the generated opinion distributions for each topic, while the red curves illustrate the corresponding approximate probability distributions. The parameter $k = 10$ indicates a moderate concentration of opinions around the corresponding mean value μ . Topic 1 exhibits an approximately right-skewed distribution, indicating that a larger proportion of agents hold relatively negative opinion values. Topic 2 displays an approximately symmetric, near-normal distribution centred around the mean $\mu = 0$. In contrast, Topic 3 exhibits an approximately left-skewed distribution, where opinions are concentrated toward more positive values.

3.2.2 Logic Matrix Construction

The number of topics is set to $m = 3$. To capture different types of belief interdependency among topics, three logic matrix structures are considered following Ye et al. (2020).

Structure (a) corresponds to a cascade structure, where earlier topics influence subsequent topics. In this case, Topic 1 depends only on itself, Topic 2 depends on Topic 1 and itself, and Topic 3 depends on Topics 1, 2, and itself.

Structure (b) represents competing interdependency, where topics exert opposing influences on each other. In this case, a topic may simultaneously receive positive and negative influences from different topics. For example, Topic 2 may be positively influenced by Topic 1 while negatively influenced by Topic 3, reflecting a direct logical conflict between beliefs.

Structure (c) represents mixed interdependency, where both reinforcing and contradictory logical relations among topics coexist. Unlike the competing structure, the positive and negative influences are determined by independent parameters, allowing greater flexibility in the strengths of reinforcing and opposing effects among topics.

The three logic matrix structures are given by

$$(a) \quad C_i = \begin{bmatrix} 1 & 0 & 0 \\ -\beta_i & 1 - \beta_i & 0 \\ \delta_i & -\eta_i & 1 - (\delta_i + \eta_i) \end{bmatrix}$$

$$\beta_i, \delta_i, \eta_i \sim \text{Beta}(2, 2), \quad \beta_i, \delta_i, \eta_i \in (0, 1)$$

$$(b) \quad C_i = \begin{bmatrix} 1 & 0 & 0 \\ \theta\beta_i & 1 - (1 + \theta)\beta_i & -\beta_i \\ \tau\delta_i & -\delta_i & 1 - \delta_i(1 + \tau) \end{bmatrix}$$

$$\beta_i, \delta_i \sim \text{Beta}(2, 2), \quad \beta_i, \delta_i \in (0, 1), \quad \theta = 0.5, \tau = 0.3$$

$$(c) \quad C_i = \begin{bmatrix} 1 & 0 & 0 \\ \eta_i & 1 - (\beta_i + \eta_i) & -\beta_i \\ \mu_i & -\delta_i & 1 - (\delta_i + \mu_i) \end{bmatrix}$$

$$\beta_i, \delta_i, \mu_i \sim \text{Beta}(2, 2), \quad \beta_i, \delta_i, \mu_i \in (0, 1)$$

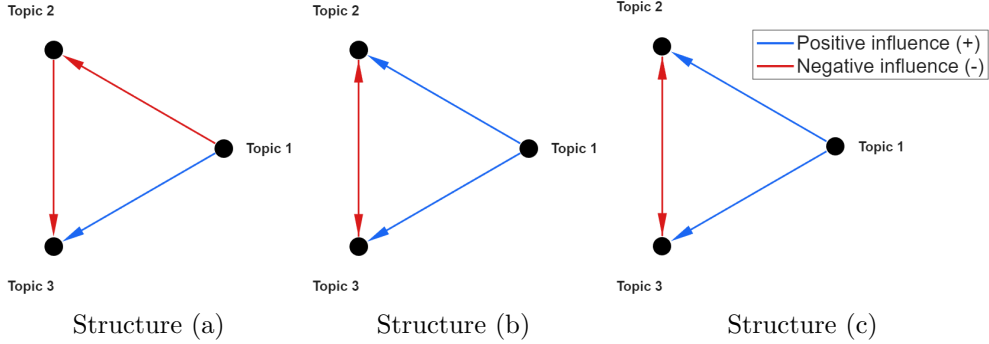


Figure 6: Graphical representation of the logic matrix structures C_i : (a) cascade structure, (b) competing interdependency, and (c) mixed interdependency.

3.2.3 Influence Network Initialisation

For the simulations, two cases are considered. In the first case, $\epsilon = 0.1$, representing agents with lower tolerance for differences in opinions, whereas in the second case, $\epsilon = 0.5$, representing agents with higher tolerance for differences in opinions. The parameter δ is set to 0.1 to allow a wider range of interaction probabilities, including the possibility of interactions occurring with relatively small probability.

Social Influence Network Construction Algorithm

Step	Description
1. Function input	Create a function that accepts $X(t)$, ϵ , and δ as inputs.
2. Initialization	Initialize an empty influence matrix W to store the influence weights w_{ij} . A helper function is used to generate the probability matrix P .
3. Updating influence weights	Loop over the n agents and update the weights w_{ij} according to the influence weight rule in Eq (2) and the stochastic interaction condition in Eq (5).
4. Row normalization	Normalize each row of W according to Eq (4) ensure that the matrix $W(t)$ is row-stochastic. Return the resulting influence matrix $W(t)$.

3.2.4 Static Social Influence Network Based on the Watts–Strogatz Model

To provide a benchmark for comparison with the time-varying social influence network, the Watts–Strogatz network generation mechanism is adopted to construct the underlying social influence structure. The mechanism follows the approach described in Gao et al. (2024). The Watts–Strogatz model is based on the concept of small-world networks, which capture the characteristics of clustering observed in many real-world social networks.

The network is generated by first forming a ring lattice with n nodes, where each node is connected to its d nearest neighbours, with $d \geq 2$ being an even integer. Subsequently, each edge is randomly rewired with probability p (Gao et al., 2024).

Watts-Strogatz (n=30, d=4, p=0.50)

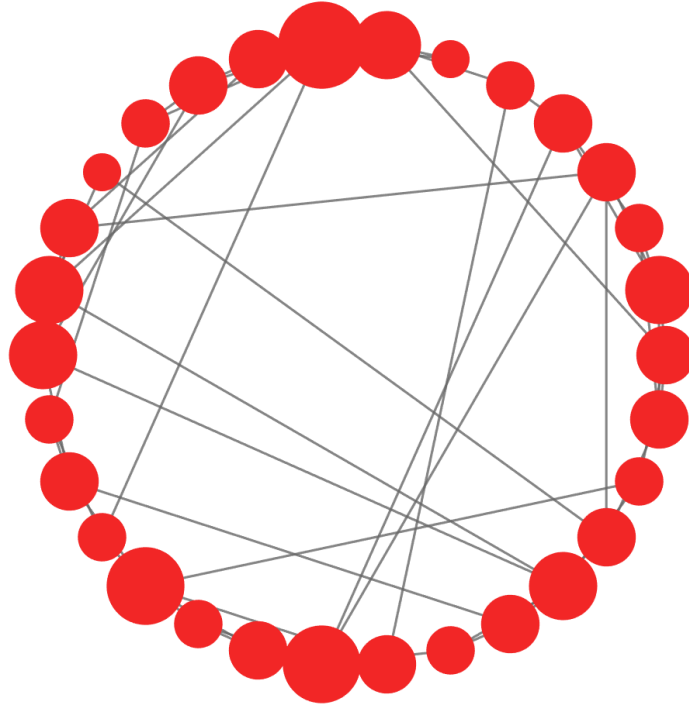


Figure 7: A static social influence network generated using the Watts–Strogatz model for $n = 30$.

Figure 7 illustrates an example of a static social influence network generated using this mechanism. d is set to 4 and p is set to 0.50. The node size corresponds to the degree of each node, representing the number of neighbouring connections in the resulting network. Nodes with more neighbours are therefore displayed with larger sizes.

3.3 Derivation of the Opinion Update Rule

This section demonstrates how the matrices defined in the previous subsection are incorporated into the computational framework to derive the opinion update rule as formulated in (2).

The social influence among agents is first incorporated as

$$\tilde{X}(t) = X(t)W(t)^\top,$$

where

$$X(t) = [x_1(t) \quad \cdots \quad x_n(t)] .$$

Substituting the matrix forms yields

$$\tilde{X}(t) = \begin{bmatrix} x_1(t) & \cdots & x_n(t) \end{bmatrix} \begin{bmatrix} w_{11}(t) & \cdots & w_{n1}(t) \\ \vdots & \ddots & \vdots \\ w_{1n}(t) & \cdots & w_{nn}(t) \end{bmatrix}.$$

Carrying out the matrix multiplication gives

$$\tilde{X}(t) = \begin{bmatrix} w_{11}(t)x_1(t) + \cdots + w_{1n}(t)x_n(t) & \cdots & w_{n1}(t)x_1(t) + \cdots + w_{nn}(t)x_n(t) \end{bmatrix}.$$

Equivalently, each column of $\tilde{X}(t)$ can be written in summation form as

$$\tilde{X}(t) = \begin{bmatrix} \sum_{j=1}^n w_{1j}(t)x_j(t) & \cdots & \sum_{j=1}^n w_{nj}(t)x_j(t) \end{bmatrix}.$$

Next, the logic matrices C_i are applied to each column of $\tilde{X}(t)$, giving

$$X(t+1) = \begin{bmatrix} C_1 \sum_{j=1}^n w_{1j}(t)x_j(t) & \cdots & C_n \sum_{j=1}^n w_{nj}(t)x_j(t) \end{bmatrix}.$$

Since C_i does not depend on j , the above expression can be rewritten as

$$X(t+1) = \begin{bmatrix} \sum_{j=1}^n w_{1j}(t)C_1x_j(t) & \cdots & \sum_{j=1}^n w_{nj}(t)C_nx_j(t) \end{bmatrix}.$$

This expression can therefore be written compactly as

$$X(t+1) = \begin{bmatrix} x_1(t+1) & \cdots & x_n(t+1) \end{bmatrix}.$$

Consequently, the i -th column of $X(t+1)$ is given by

$$x_i(t+1) = \sum_{j=1}^n w_{ij}(t)C_ix_j(t).$$

Overall, the derivation above corresponds to the MATLAB computational implementation.

Based on the model initialisation, logic matrix structures, and influence network construction described in previous subsections, the complete simulation procedure is summarised in the following algorithm.

Opinion Dynamics Simulation Algorithm

Step	Description
1. Function input	Create a function that accepts $X(0)$, C , T , ϵ , and δ as inputs. The variable C stores the set of logic matrices C_i , and T represents the simulation duration.
2. Initialization	Initialize an opinion history matrix to store the opinions of all agents at each time step.
3. Time evolution	Loop from $t = 1$ to T to simulate the opinion dynamics over time.
4. Influence network creation	At each time step t , construct the influence matrix $W(t)$ using a helper function that takes $X(t)$, ϵ , δ , and t as inputs.
5. Social influence update	Compute the intermediate opinion matrix $\tilde{X}(t) = X(t)W(t)^\top$.
6. Logical update	Loop over agents $i = 1, \dots, n$ and update the opinion vector using the corresponding logic matrix C_i .
7. Record state	Store the updated opinion matrix $X(t+1)$ in the opinion history matrix and return the opinion trajectory matrix <i>opinion_history</i> .

4 Results

To evaluate the proposed opinion dynamics model with time-varying influence weights, simulation experiments are conducted under three different network settings. These include a static social influence network and the proposed time-varying networks with bounded confidence thresholds $\epsilon = 0.5$ and $\epsilon = 0.1$. For each network setting, three logic interdependency structures are considered: cascade, competing, and mixed structures.

4.1 Static Social Influence Network

Under the constant social influence network, the influence matrix remains fixed throughout the simulation. This setting serves as a baseline for comparison with the proposed time-varying network mechanism.

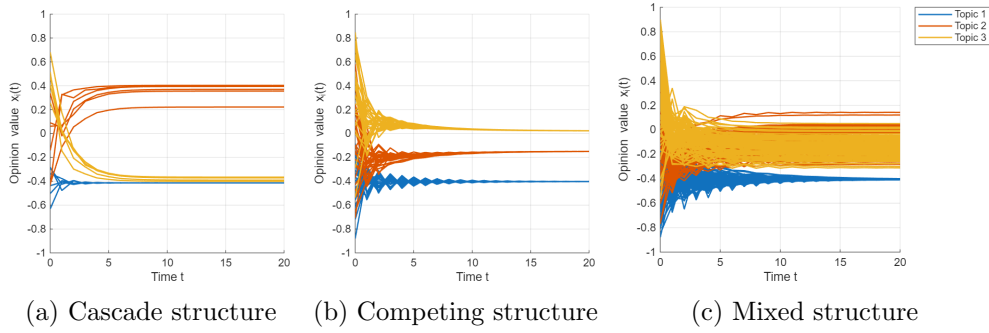


Figure 8: Opinion dynamics under the static social influence network for three logic structures. Panels (a)–(c) correspond to cascade, competing, and mixed interdependency, respectively. The number of agents used in the simulations are $n = 6$, $n = 30$, and $n = 100$ for panels (a), (b), and (c), respectively. In panel (a), one agent is assigned a non-negative value for the matrix entry $C_{2,1}$ while the remaining agents follow the standard cascade structure.

The figures illustrate the evolution of agents' opinions over time for different logic structures. Each curve represents the opinion value of an individual agent on a particular topic, while the horizontal axis denotes the time step and the vertical axis shows the corresponding opinion value. At time step $t = 0$, agents begin with different initial opinion values, representing varying degrees of agreement and disagreement across the population. As time progresses, agents interact with one another through the social influence network, allowing opinions to propagate across the network. Through these interactions, consensus, diversity, or clustering may emerge depending on the mechanism that governs it.

The results show that the opinion values of agents gradually evolve and reach a steady state around time step 15. These outcomes are consistent with the results reported in Ye et al. (2020) where the cascade structure with heterogeneity and mixed interdependency structure lead to opinion diversity, whereas the competing structure gives rise to consensus for each topic. In this simulation, the underlying social influence network is strongly connected, ensuring that opinion information can propagate throughout the entire network.

In Figure 8(a), the introduction of heterogeneity—where one agent is assigned a different C_i logic structure—results in diversity in the final opinion values. In contrast, the competing logical interdependency leads to consensus for each topic as observed by the behaviour in Figure 8(b). Meanwhile, the mixed logical interdependency in Figure 8(c) again produces diversity in the final opinion values. These results highlight that the structure of the logic matrices plays an important role in determining the steady-state opinion configuration.

4.2 Time-Varying Social Influence Network with $\epsilon = 0.5$

In this setting, the social influence network evolves over time according to the proposed influence weight formulation. The bounded confidence threshold is set to $\epsilon = 0.5$, indicating a relatively high tolerance among agents toward opinion dissimilarity. Consequently, agents are able to interact with others whose opinions differ within a broader range.

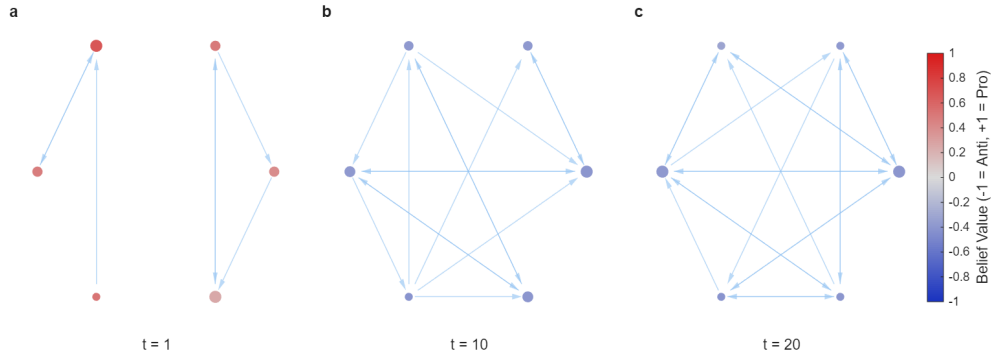


Figure 9: Time-Varying Social Influence Network with $\epsilon = 0.5$.

Figure 9 illustrates the evolution of a social influence network consisting of $n = 6$ agents. Initially, the network is weakly connected because the disparity in the initial opinion distribution exceeds the bounded confidence threshold. As agents repeatedly interact, their opinion values gradually become closer and fall within the bounded confidence range. Consequently, the network progressively evolves and eventually becomes strongly connected.

Due to the stochastic interactions occurring at each time step, a different strongly connected network may be formed at every time t . Despite this variability, the time-varying networks produce results that are generally similar to those obtained under the static network setting. However, when consensus emerges, it is typically reached faster in the time-varying network. This may be explained by the fact that the changing strongly connected structures and homophily allow information to propagate more efficiently across the network.

For example, as illustrated in Figure 10, in a static network, agent 3 can only receive influence indirectly from agent 1 through agent 2. In contrast, in a time-varying network, at time step $t = 5$, agent 3 is able to receive direct influence from agent 1. In a large population such as $n = 100$, where opinions are moderately concentrated, influence weights are determined based on opinion similarity. As opinions become increasingly similar, more connections are formed within the network, enabling more

efficient propagation of information.

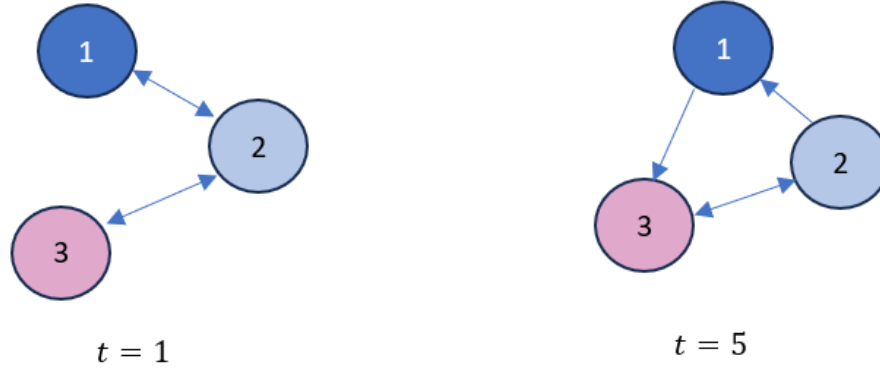


Figure 10: Time-Varying Social Influence Network with $\epsilon = 0.5$.

However, when diversity emerges, the opinion trajectories for Topics 2 and 3 exhibit oscillatory behaviour in steady state, as illustrated in Figure 11(a).

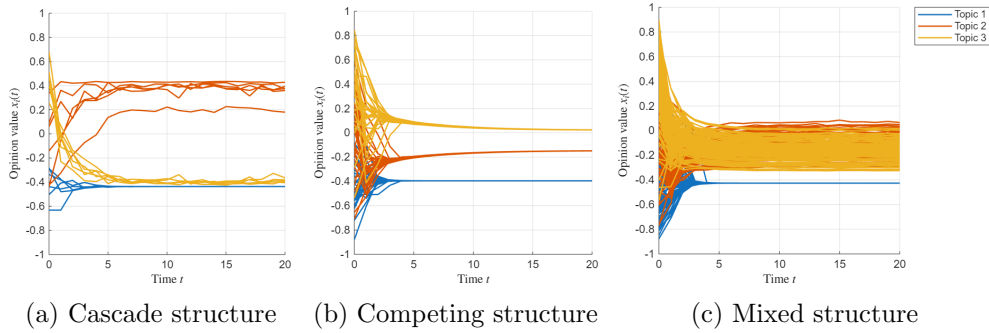


Figure 11: Opinion dynamics under a time-varying social influence network with $\epsilon = 0.5$ for three logic structures. Panels (a)–(c) correspond to cascade, competing, and mixed interdependency, respectively. The number of agents used in the simulations are $n = 6$, $n = 30$, and $n = 100$ for panels (a), (b), and (c), respectively. In panel (a), one agent is assigned a non-negative value for the matrix entry $C_{2,1}$ while the remaining agents follow the standard cascade structure.

The fluctuations are primarily caused by the stochastic interaction mechanism, which results in each agent having a different set of connections at each time step. Consequently, the number of neighbours influencing an agent varies over time, leading to small variations in the

opinion updates according to the update rule

$$x_i(t+1) = \sum_{j=1}^n w_{ij}(t) C_i x_j(t)$$

For example, at time step $t = 15$, agent i may receive influence from agents $j = \{1, 3\}$, whereas at time step $t = 16$, the set of influencing agents may change to $j = \{2, 4, 5\}$. This variation in the interaction structure causes the opinion trajectories to fluctuate over time.

To further investigate this effect, the stochastic interaction mechanism is removed while keeping the time-varying network structure. In this case, the fluctuations become much more subtle and almost disappear, as shown in Figure 12. This indicates that the oscillatory behaviour observed in the opinion trajectories is largely driven by the random interaction mechanism.

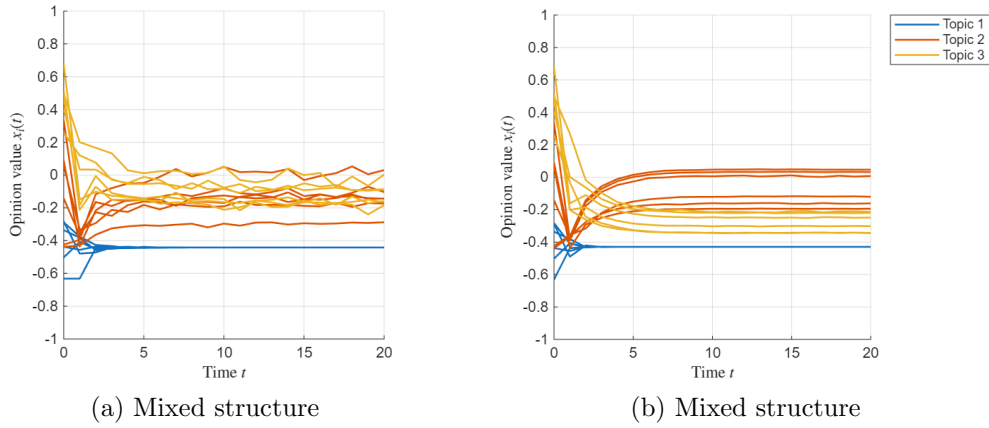


Figure 12: Opinion dynamics under a time-varying social influence network with $\epsilon = 0.5$ for a mixed interdependency structure. Panels (a) and (b) illustrate the time-varying network with stochastic interaction and without stochastic interaction, respectively. The number of agents used in the simulation is $n = 6$.

4.3 Time-Varying Social Influence Network with $\epsilon = 0.1$

When the bounded confidence threshold is reduced to $\epsilon = 0.1$, interactions among agents become more restricted. Agents only interact with others whose opinions are sufficiently similar, resulting in a more fragmented influence network as shown in Figure 13

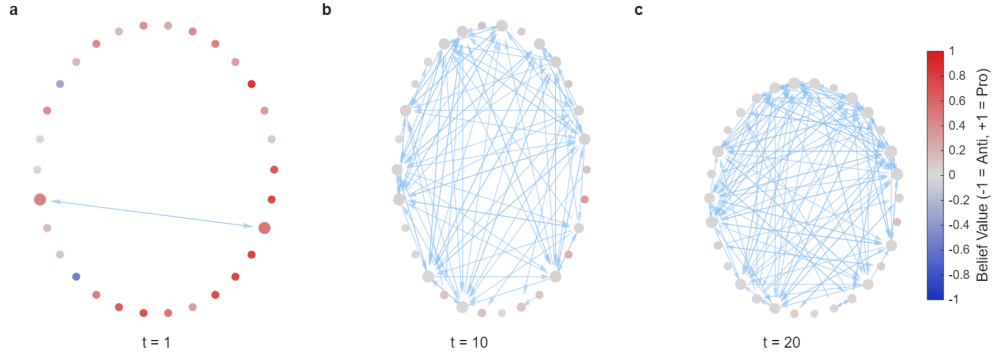


Figure 13: Time-Varying Social Influence Network with $\epsilon = 0.5$.

Under this setting, strong diversity and fragmentation in opinions are observed in the steady-state opinion values, resulting in high variance across agents as there is a combination of disconnected component, weakly connected component, and strongly connected component in the network. The presence of strongly connected components within the social influence network results in clusters of agents with similar steady-state opinions as shown in Figure 14.

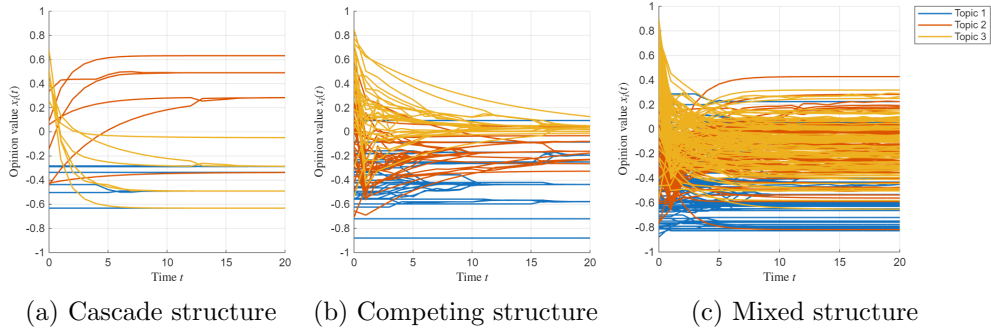


Figure 14: Opinion dynamics under a time-varying social influence network with $\epsilon = 0.1$ for three logic structures. Panels (a)–(c) correspond to cascade, competing, and mixed interdependency, respectively. The number of agents used in the simulations are $n = 6$, $n = 30$, and $n = 100$ for panels (a), (b), and (c), respectively. In panel (a), one agent is assigned a non-negative value for the matrix entry $C_{2,1}$ while the remaining agents follow the standard cascade structure.

Depending on the logic structure, the mean of the steady-state opinion values may remain close to their initial values, gravitate towards another topic, or diverge from it. In the mixed interdependency structure, Topic 1 does not depend on any other topics; hence its steady-state mean opinion value remains relatively close to its initial mean. In contrast, the steady-state mean opinion values of Topics 2 and 3 tend to

gravitate towards Topic 3.

Although this behaviour is not very apparent in Figures 8 and 11, the variances of Topics 2 and 3 in a time-varying social influence network with $\epsilon = 0.5$ are slightly lower than those in the static network. However, when $\epsilon = 0.1$, the variances in the time-varying social influence network become substantially larger compared with the other network settings. Table 1 presents the mean and variance for the different network settings when $n = 1000$ agents.

Table 1: Mean and variance of steady-state opinion values across topics for a static network and time-varying networks with $\epsilon = 0.5$ and $\epsilon = 0.1$ at time $t = 20$ for $n=1000$

Network Type	Topic	Mean	Variance
Static network	Topic 1	-0.39	0
	Topic 2	-0.13	0.0063
	Topic 3	-0.13	0.0058
Time-varying network ($\epsilon = 0.5$)	Topic 1	-0.40	0
	Topic 2	-0.12	0.0047
	Topic 3	-0.15	0.0046
Time-varying network ($\epsilon = 0.1$)	Topic 1	-0.39	0.0687
	Topic 2	-0.11	0.0301
	Topic 3	-0.13	0.0331

4.4 Discussion

The simulation results provide clear insights into the types of social influence network structures that emerge under homophily with stochastic interactions. The network structure, governed primarily by the homophily parameter ϵ , determines whether agents can exchange information globally or remain confined within local clusters. When the confidence threshold is relatively large ($\epsilon = 0.5$), agents are able to interact with a broader set of neighbours, allowing initially weakly connected networks to evolve into strongly connected structures over time which promote connectivity. In contrast, when the confidence threshold is small ($\epsilon = 0.1$), interactions become highly selective due to homophily. As a result, the network remains fragmented, consisting of a mixture of disconnected, weakly connected, and strongly connected components. This fragmentation restricts the flow of information across the network and leads to the formation of multiple opinion clusters, opinion fragmentation and persistent diversity in opinions. Furthermore, the presence of stochas-

tic interactions introduces temporal variability in the network structure. The findings from both network structures reflect the dynamic nature of real-world social influence networks, in which interpersonal connections may strengthen, weaken, emerge, or disappear over time, leading to the continuous evolution of the network structure.

When the network becomes strongly connected, the specific interaction patterns vary over time, resulting in dynamic changes in connectivity. This contributes to the observed oscillatory behaviour in opinion trajectories. The findings reflect real-world opinion dynamics in society, where opinions are subject to continual change and fluctuations over time.

In addition, the logical interdependency structure encoded in the matrices C_i significantly influences the steady-state outcomes. Competing logical structures tend to drive opinions toward consensus for each topic, whereas cascade structure with introduction of heterogeneity and mixed structures can sustain diversity. The findings contribute to the understanding of how consensus and diversity of opinions arise within a strongly connected society.

Nevertheless, the model has several limitations. First, the current work primarily explores weakly connected structures of the logic matrices. Future studies could investigate belief dynamics under strongly connected logic structures to better understand how tightly coupled topic dependencies influence opinion outcomes. Second, the social influence weights in this study are derived from a homophily-based distance function. Future work could explore alternative weighting functions derived from empirical observations of real social interactions. Finally, the model assumes a homogeneous bounded-confidence ϵ , whereas real individuals may possess different tolerance levels toward opinion differences. Future research could therefore consider heterogeneous ϵ values across agents to better capture realistic social behaviour.

5 Conclusion

This study investigates belief dynamics in time-varying social influence networks with logic constraints and stochastic interactions, whereas existing literature typically considers only some of these aspects in isolation. For instance, the classical DeGroot model assumes a static social influence network with only a single topic, while the Hegselmann–Krause (HK) model considers only a single topic without logical constraints between topics. The proposed model extends and synthesises previous agent-based opinion dynamics models by incorporating time-varying social influence networks, logically interdependent topics, and stochastic

interactions, together with a novel mechanism for calculating influence weights based on opinion similarity.

The results show that homophily governs the evolution of the network structure over time. In particular, there exists a minimum bounded confidence threshold ϵ above which the network gradually evolves into a strongly connected structure as agents repeatedly interact and their opinions become more similar.

The structure of the resulting social influence network strongly influences the diversity of the final opinions. When the network contains disconnected, weakly connected, or strongly connected components simultaneously, the opinion dynamics tend to produce strong diversity in the steady-state opinion values. This occurs because information cannot fully propagate across all agents, resulting in clusters of agents converging to different opinions.

The results also indicate that stochastic interactions are the primary driver of oscillatory behaviour in opinion trajectories. Since the interaction structure changes at each time step, the set of neighbours influencing an agent varies over time, which introduces fluctuations in the opinion updating process. When the stochastic interaction mechanism is removed, these oscillations become significantly weaker.

In addition, when the social network is strongly connected, the logic structure plays an important role in determining whether opinions converge or remain diverse. In particular, the interdependency among topics determines the direction in which opinions evolve and whether consensus is achieved. Diversity in logic structures across agents further contributes to heterogeneous opinion outcomes.

The findings highlight the importance of the interplay between homophily, stochastic interaction, and logical interdependency in shaping belief dynamics in evolving social influence networks. However, further extensions can be undertaken to investigate strongly connected logic matrices and heterogeneous bounded-confidence thresholds among agents. In addition, alternative influence-weight calculation functions may be explored to examine the robustness of the results under different interaction mechanisms.

Acknowledgements

I would like to express my deepest and most sincere gratitude to my project supervisor, Dr Mengbin Ye, for his patience, time, and guidance throughout my research journey. His constructive feedback and expert insights consistently guided me in the right direction and provided valuable ideas for improving my project. The successful completion of this research would not have been possible without his encouragement to think critically and maintain a high standard of analysis. I am truly grateful for his continuous support and dedication.

I would also like to acknowledge Yongwen Pan for providing the original code used to generate the opinion matrix in my simulation.³ I also adapted code from Pui Hang Wong for plotting snapshots of the social influence network at three different time steps.⁴ In particular, I adapted the plotting approach by fixing all nodes in a lattice-ring layout and adding arrows to represent the directed nature of the graph.

In addition, I acknowledge the use of ChatGPT [<https://chat.openai.com/>] as an AI-assisted tool for guidance in coding by providing step-by-step coding support, syntax guidance and explanations. Below are some of the prompts that were entered into ChatGPT approximately on 30th October 2025, 13th November 2025, 20th November 2025, and 26th March 2026:

- How can I create a MATLAB function that returns the variable 'opinion_history' as an output and accepts X , C , and T as inputs
- How can I create a three-dimensional matrix with dimensions (m, n, T)
- Show me an example of matrix multiplication
- How can I update and store results in a matrix
- How can I add legends with different colours for each plots
- Show me an example of how to calculate mean and variance from a three-dimensional matrix.

I acknowledge the use of ChatGPT [<https://chat.openai.com/>] to identify suitable packages for editing my report template in Overleaf and to convert mathematical expressions and equations written in Microsoft

³The original code for generating the X matrix can be found at: https://github.com/yongwenpoon/FJ_Opinion_Generator.

⁴The original code for plotting the social influence network can be found at: <https://github.com/viviwph/belief-dynamics-platform/tree/viz>.

Word into latex format for usage in Overleaf on 28th April 2026 and 1st May 2026.

I acknowledge the use of ChatGPT [<https://chat.openai.com/>] to suggest more academic-sounding terms and refine my grammar in the report on 2nd May 2026 and 7 May 2026. It did not generate new contents or arguments.

References

- Brooks, H. Z., & Porter, M. A. (2020). A model for the influence of media on the ideology of content in online social networks. *Physical Review Research*, 2(2).
- Cheng, J., Chen, G., Mei, W., & Bullo, F. (2025). Multidimensional opinion dynamics with heterogeneous bounded confidences and random interactions. *Automatica*, 172, 112002. <https://doi.org/10.1016/j.automatica.2024.112002>
- Converse, P. E. (2008). The nature of belief systems in mass publics (1964). *Critical Review*, 18(1-3), 1–74. <https://doi.org/10.1080/08913810608443650>
- Degroot, M. H. (1974). Reaching a consensus. *Journal of the American Statistical Association*, 69(345), 118–121. <https://doi.org/10.1080/01621459.1974.10480137>
- Friedkin, N. E., Proskurnikov, A. V., Tempo, R., & Parsegov, S. E. (2016). Network science on belief system dynamics under logic constraints. *Science*, 354(6310), 321–326. <https://doi.org/10.1126/science.aag2624>
- Gao, T., Zino, L., & Ye, M. (2024). Effect of network structure and committed minority placement in promoting social diffusion. *IEEE Transactions on Computational Social Systems*, 11(2). <https://doi.org/10.1109/TCSS.2023.3303568>
- Hegselmann, R., & Krause, U. (2002). Opinion dynamics and bounded confidence models, analysis, and simulation. *Journal of Artificial Societies and Social Simulation*, 5(3). <http://jasss.soc.surrey.ac.uk/5/3/2.html>
- Kramer, G. H. (1971). Short-term fluctuations in u.s. voting behavior, 1896–1964. *American Political Science Review*, 65(1), 131–143. <https://doi.org/10.2307/1955049>
- León-Medina, F. J. (2019). Endogenous changes in public opinion dynamics. *Journal of Artificial Societies and Social Simulation*, 22(2), 4. <https://doi.org/10.18564/jasss.3967>
- McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27(1), 415–444. <https://doi.org/10.1146/annurev.soc.27.1.415>
- Usó-Doménech, J. L., & Nescolarde-Selva, J. (2015). What are belief systems? *Foundations of Science*, 21(1), 147–152. <https://doi.org/10.1007/s10699-015-9409-z>
- Ye, M., Liu, J., Wang, L., Anderson, B. D. O., & Cao, M. (2020). Consensus and disagreement of heterogeneous belief systems in influ-

ence networks. *IEEE Transactions on Automatic Control*, 65(11), 4679–4694. <https://doi.org/10.1109/TAC.2019.2961998>